

Here are the detailed instructions for your Power BI homework tasks, which you can copy and paste into a document to save as a PDF.

Power BI Homework: Working with Maps & Geocoding, Formatting & Annotations

Prerequisites: Download the `sales_with_geodata.csv` file.

Part 1: Maps & Geocoding

Task 1: Create a Basic Map Showing Sales by Country

1. **Import the Data:**
 - Open Power BI Desktop.
 - Go to the "Home" tab and click "Get data."
 - Select "Text/CSV" and import the `sales_with_geodata.csv` file.
 - Review the data in the preview window and click "Load."
2. **Add the Map Visual:**
 - From the "Visualizations" pane, select the "Map" visual (the icon with a globe).
3. **Configure the Visual:**
 - Drag the `Country` field from the "Data" pane to the **Location** field well of the "Map" visual. Power BI will automatically recognize this as a geographical location.
 - Drag the `Total Sales` field to the **Bubble size** field well.
4. **Result:** The map will display bubbles over each country, with the size of the bubble proportional to the total sales in that country.

Task 2: Use Latitude and Longitude for Precise Mapping

1. **Duplicate the Map Visual:**
 - Right-click the map visual you created in Task 1 and select "Copy."
 - Paste it onto the report canvas.
2. **Remove the Country Field:**
 - Remove `Country` from the **Location** field well of the new map visual.
3. **Add Latitude and Longitude:**
 - Drag the `Latitude` field to the **Latitude** field well.
 - Drag the `Longitude` field to the **Longitude** field well.
 - Make sure the summarization for both `Latitude` and `Longitude` is set to "Don't summarize" by clicking the dropdown arrow next to the field name in the well.
4. **Add Sales Data:**
 - Drag the `Total Sales` field to the **Bubble size** field well.
5. **Result:** The map will now show individual data points for each specific location (based on the provided coordinates), with the bubble size representing the sales at that exact point.

Task 3: Plot Total Sales by City

1. **Create a New Map Visual:**
 - Add a new "Map" visual to the report canvas.
2. **Configure the Visual:**
 - Drag the **City** field to the **Location** field well.
 - Drag the **Total Sales** field to the **Bubble size** field well.
3. **Result:** The map will display a bubble for each city, with the size corresponding to the total sales in that city. Power BI's geocoding will automatically attempt to locate the cities.

Task 4: Create a Region-wise Sales Heatmap (Use the Shape Map visual)

1. **Add the Shape Map Visual:**
 - From the "Visualizations" pane, select the "Shape Map" visual. (If you don't see it, you may need to enable it in **File > Options and settings > Options > Preview features**).
2. **Configure the Visual:**
 - Drag the **Country** field to the **Location** field well.
 - Drag the **Total Sales** field to the **Color saturation** field well.
3. **Select a Map:**
 - In the "Format your visual" pane, go to **Map settings**.
 - From the "Map" dropdown, select a map that corresponds to your data (e.g., "World Map" or a specific country map if your data is localized). Power BI will shade the countries based on **Total Sales**.
4. **Result:** The countries on the map will be colored with a gradient, where a darker or more intense color indicates higher total sales, creating a heatmap effect.

Task 5: Add Map Drill-Down from Country → Region → City

1. **Create a Hierarchy:**
 - In the "Data" pane, right-click the **Country** field.
 - Select "Create hierarchy." This will create a **Country Hierarchy**.
 - Drag the **Region** field into the newly created **Country Hierarchy**.
 - Drag the **City** field into the **Country Hierarchy**. Your hierarchy should now be **Country > Region > City**.
2. **Configure the Map Visual:**
 - Add a new "Map" visual.
 - Drag the **Country Hierarchy** you just created to the **Location** field well.
 - Drag **Total Sales** to the **Bubble size** field well.
3. **Use the Drill-Down Buttons:**
 - In the top-left corner of the map visual, you will see a set of drill-down icons (down arrow, double down arrow, fork icon).

- Click the single down arrow to drill down from **Country** to **Region**, and click it again to drill down to **City**. The map will update to show bubbles for the new level of the hierarchy.

Part 2: Formatting & Annotations

Task 6: Customize Map Visual – Titles, Colors, and Labels

1. **Select the Map Visual:**
 - Click on one of the map visuals you created (e.g., the one from Task 1).
2. **Open the Format Pane:**
 - Go to the "Format your visual" pane (the paintbrush icon).
3. **Customize Title:**
 - Expand the **General** section, then expand **Title**.
 - Change the "Title" text to something descriptive like "Total Sales by Country."
 - Adjust the text color, size, and alignment.
4. **Customize Bubble Colors:**
 - Expand the **Visual** section, then expand **Bubbles**.
 - Change the **Color** to your preferred color. You can also adjust the "Size" to make the bubbles smaller or larger.
5. **Add Labels:**
 - Expand the **Map settings** section.
 - Under "Map style," you can change the visual style of the map itself (e.g., Aerial, Dark, Light, etc.).
 - Expand the **Visual** section, then **Labels**.
 - Turn on the "Category labels" to show the names of the locations on the map.

Task 7: Use Conditional Formatting for Bubble Colors

1. **Select the Map Visual:**
 - Click on the map visual from Task 3 (Total Sales by City).
2. **Access Conditional Formatting:**
 - In the "Visualizations" pane, in the **Bubbles** field well, click the dropdown arrow next to **Total Sales**.
 - Select "Conditional formatting" and then choose "Field value." (**Note:** For bubble color, you typically do this in the Format pane, not the field well.)
 - Alternatively, in the "Format your visual" pane, go to **Visual > Bubbles > Colors**, and click the "fx" button.
3. **Set Formatting Rules:**
 - A "Conditional formatting" window will pop up.
 - Select "Gradient" from the "Format by" dropdown.
 - Select **Total Sales** as the "Based on field."

- Choose a color scale. For example, set the "Minimum" color to a light color (e.g., yellow) and the "Maximum" color to a dark color (e.g., dark blue).
 - Click "OK."
4. **Result:** The bubbles on the map will now be colored on a gradient scale based on their **Total Sales**, providing a quick visual reference for performance.

Task 8: Add Annotations to Highlight Key Cities

1. **Select the Map Visual:**
 - Choose the map visual from Task 3 (Total Sales by City).
2. **Add a Comment:**
 - Hover over a specific city bubble on the map that you want to highlight.
 - A small tooltip will appear. You may need to have "Comments" enabled in the "View" tab.
 - Click the "Comment" icon (a speech bubble).
 - Type your annotation text, such as "Highest Sales in this Region!" or "New Market."
 - This feature works best when the report is published to the Power BI Service, but you can also use Shapes and Text Boxes on the report canvas to create a similar effect.
3. **Alternative Method (Shapes & Text Boxes):**
 - Go to the "Insert" tab in the Power BI ribbon.
 - Select "Shapes" and choose an arrow or a line.
 - Drag the shape to point at the city on the map.
 - Select "Text box" and type your annotation. Place it near the arrow.

Task 9: Create a Tooltip Page with Detailed Info

1. **Create a New Report Page:**
 - Click the **+** button at the bottom of the canvas to create a new page.
 - Rename the page to "City Details Tooltip."
2. **Configure the Page as a Tooltip:**
 - In the "Visualizations" pane, go to the "Format your page" section.
 - Expand the "Page information" section and turn "Tooltip" on.
 - Expand the "Page size" section and change the "Type" to "Tooltip."
3. **Add Visuals to the Tooltip Page:**
 - Add a visual that shows detailed information. For example, a card visual showing **Total Sales**, a bar chart showing sales by **Category**, and a table showing **Product Name** and **Quantity**.
4. **Assign the Tooltip Page to the Map:**
 - Go back to your map visual from Task 3.
 - In the "Format your visual" pane, go to **General > Tooltips**.
 - Under "Type," select "Report page."
 - Under "Page," select the "City Details Tooltip" page you just created.

5. **Result:** Now, when you hover over any city bubble on the map, a custom tooltip will appear with the detailed information you placed on the tooltip page.

Task 10: Add a Slicer to Filter the Map by Category

1. **Add a Slicer Visual:**
 - From the "Visualizations" pane, select the "Slicer" visual.
 2. **Configure the Slicer:**
 - Drag the **Category** field to the **Field** well of the slicer visual.
 3. **Connect to the Map:**
 - Power BI slicers automatically filter all related visuals on the same page. The map will be filtered by default.
 4. **Result:** You can now click on different categories in the slicer (e.g., "Electronics," "Apparel") to see how the sales on your map change for that specific category.
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